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OpenGeoSys Training Course

Coupled Processes in Geological Media with OpenGeoSys

Course exercises: tunnel excavation (Kirsch benchmark), fluid injection into fractured rock (LIE vs. EFPM), and thermo-hydro-mechanics of a heated tunnel (TRM)

Mehran Ghasabeh, Thomas Nagel

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Institute of Geotechnics, TU Bergakademie Freiberg

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1 Introduction

This report documents the three benchmark exercises of the joint OpenGeoSys training course. All exercises use the open-source finite-element simulator OpenGeoSys (OGS) [3, 4, 8], version 6.5.9, installed as a Python wheel. The exercises increase in physical complexity:

1. **Tunnel excavation – Kirsch benchmark** (`Tunnel_Excavation_Kirsch/`): linear-elastic excavation of a circular tunnel with the `ReleaseNodalForce` boundary condition, verified against the analytical Kirsch solution [1].
2. **Fluid injection into fractured rock** (`LIE_EFPM/`): hydro-mechanically coupled injection into a fractured, nearly impermeable matrix, comparing lower-dimensional interface elements (LIE) with the embedded fracture permeability model (EFPM), verified against analytical and one-dimensional finite-difference references.
3. **Heated tunnel in Opalinus Clay** (`TRM/`): coupled thermo–hydro–mechanical response of a bentonite-buffered, heat-emitting tunnel modelled with the `THERMO_RICHARDS_MECHANICS` process, following the heater benchmark of Pitz et al. [2] (DECOVALEX-2023 Task C), in a fully saturated and a partially saturated configuration.

1.1 Common workflow

All three exercises follow the same reproducible structure:

- A single Jupyter notebook per problem — `01_tunnel_excavation_kirsch.ipynb`, `02_lie_efpm_injection.ipynb`, and `03_trm_heated_tunnel.ipynb` — is the only place where parameter values are defined. It drives the complete workflow: meshing, project-file generation, OGS execution, post-processing, and verification.
- A mesh module (`mesh_*.py`, class `MeshGenerator`) builds structured Gmsh meshes, converts them to OGS `vtu` format, and extracts the boundary meshes.
- A project module (`prj_*.py`, class `PrjGenerator` or `ProjectGenerator`) assembles the OGS project file with the `ogstools Project` API. All physical and numerical settings enter as required keyword arguments; the modules contain no hidden defaults.

This separation—*values in the notebook, structure in small Python modules, verification against analytical or published references*—is intended as a reusable template for the participants’ own models.

1.2 Installation

The course was developed and tested on Linux with Python 3.12; macOS and Windows work analogously. All software installs into one virtual environment; no compilation of OGS is required:

```
python3 -m venv ~/.venvs/ogs312
source ~/.venvs/ogs312/bin/activate
pip install --upgrade pip
pip install ogs==6.5.9 ogstools==0.8.2 gmsh pyvista jupyter
python -m ipykernel install --user --name ogs312 --display-name "Python (ogs312)"
```

Open any course notebook in Jupyter and select the `Python (ogs312)` kernel. Running all cells regenerates meshes and project files, runs OGS, and produces the figures. Separate step-by-step installation guides for Windows, macOS, and Ubuntu are provided alongside this report.

One known limitation is relevant for the second exercise: in released OGS versions, including the 6.5.9 wheel, the process `HYDRO_MECHANICS_WITH_LIE` supports only a *single* displacement jump. The double-fracture case is therefore run with the EFPM approach; double-fracture LIE simulations require a custom build from the development branch `LIE_HM_multipleDisplacementJump` and are not part of the course.

2 Problem 1: Tunnel excavation and the Kirsch benchmark

2.1 Purpose

Excavation removes both material and the tractions that this material exerted on the surrounding rock. The OGS `ReleaseNodalForce` boundary condition represents this unloading while retaining a mesh that already contains the opening: the model begins from an initially stressed, undeformed state and progressively withdraws the equivalent nodal forces on the excavation boundary. After working through this exercise a participant should be able to

- distinguish an initial-stress excavation analysis from a model loaded from an unstressed reference configuration;
- derive the non-equilibrium initial residual and the equivalent nodal forces on the excavation wall;
- construct the plane-strain initial stress consistently; and
- verify OGS results against the Kirsch solution without confusing discretization error with nodal stress-extrapolation artifacts.

A dedicated companion report with the complete derivation is provided in the folder `Tunnel_Excavation_Kirsch/report/`; this chapter summarizes it.

2.2 Benchmark definition

A quarter model of a circular opening is used; the origin of the polar system is the tunnel centre and θ is measured counter-clockwise from the positive horizontal axis. Gravity is disabled; the in-situ stress is prescribed directly.

Quantity	Symbol	Value
Tunnel radius	a	6.5 m
Quarter-domain extent	L	70 m
Young's modulus	E	5.0 GPa
Poisson's ratio	ν	0.30
Vertical compressive stress magnitude	P	20 MPa
Excavation duration	t_e	$1.728 \times 10^5 \text{ s} = 2 \text{ d}$
Simulation end time	t_{end}	$3.456 \times 10^5 \text{ s} = 4 \text{ d}$

OGS uses a tension-positive stress convention, so the applied far-field stress is $\sigma_\infty = -P = -20 \text{ MPa}$. Symmetry conditions are $u_x = 0$ on the left and $u_y = 0$ on the bottom boundary; a vertical traction $\bar{t}_y = -P$ acts on the top; the right boundary is traction-free, consistent with the remote uniaxial stress state. The finite outer boundary lies at $L/a \approx 10.8$, so small differences from the infinite-domain analytical solution are expected near the outer boundary.

2.2.1 Plane-strain initial stress

The two-dimensional small-deformation process stores the stress vector as $\boldsymbol{\sigma} = [\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{xy}]^T$. Plane strain requires $\varepsilon_{zz} = 0$, and isotropic Hooke's law then yields the closure

$$\sigma_{zz} = \nu (\sigma_{xx} + \sigma_{yy}). \quad (2.1)$$

For the intact uniaxial initial state, $\sigma_{0,xx} = 0$ and $\sigma_{0,yy} = -P$, hence

$$\sigma_{0,zz} = -\nu P = -6 \text{ MPa}, \quad \boldsymbol{\sigma}_0 = [0, -20, -6, 0]^T \text{ MPa}. \quad (2.2)$$

Omitting $\sigma_{0,zz}$ does not change the in-plane equilibrium equations, but it shifts the computed σ_{zz} by 6 MPa and therefore corrupts comparisons of the full stress tensor, the mean stress, and the octahedral shear stress.

2.3 Analytical Kirsch solution

For a traction-free circular opening of radius a in an infinite isotropic elastic medium under the remote vertical stress σ_∞ , the in-plane stresses [1] are

$$\sigma_{rr}(r, \theta) = \frac{\sigma_\infty}{2} \left(1 - \frac{a^2}{r^2} \right) - \frac{\sigma_\infty}{2} \left(1 - \frac{4a^2}{r^2} + \frac{3a^4}{r^4} \right) \cos 2\theta, \quad (2.3)$$

$$\sigma_{\theta\theta}(r, \theta) = \frac{\sigma_\infty}{2} \left(1 + \frac{a^2}{r^2} \right) + \frac{\sigma_\infty}{2} \left(1 + \frac{3a^4}{r^4} \right) \cos 2\theta, \quad (2.4)$$

$$\sigma_{r\theta}(r, \theta) = \frac{\sigma_\infty}{2} \left(1 + \frac{2a^2}{r^2} - \frac{3a^4}{r^4} \right) \sin 2\theta. \quad (2.5)$$

At $r = a$: $\sigma_{rr} = \sigma_{r\theta} = 0$ (traction-free wall), $\sigma_{\theta\theta} = 3\sigma_\infty = -60 \text{ MPa}$ at the sidewall ($\theta = 0$), and $\sigma_{\theta\theta} = -\sigma_\infty = +20 \text{ MPa}$ at the crown ($\theta = 90^\circ$). The compatible out-of-plane stress follows from Eq. (2.1) applied to the rotation-invariant in-plane trace:

$$\sigma_{zz}(r, \theta) = \nu [\sigma_{rr}(r, \theta) + \sigma_{\theta\theta}(r, \theta)]. \quad (2.6)$$

2.4 Release-nodal-force formulation

For quasi-static small deformation, the discrete residual of the weak equilibrium form reads

$$\mathbf{r}(\hat{\mathbf{u}}, t) = \mathbf{F}^{\text{int}}(\hat{\mathbf{u}}) - \mathbf{F}^{\text{ext}}(t) = \mathbf{0}, \quad \mathbf{F}^{\text{int}} = \int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma} \, d\Omega, \quad \boldsymbol{\sigma} = \boldsymbol{\sigma}_0 + \mathbb{C} : \boldsymbol{\varepsilon}(\mathbf{u}). \quad (2.7)$$

At $t = 0$ the mesh already contains the opening, carries the intact in-situ stress $\boldsymbol{\sigma}_0$, and has zero displacement. This state is not in equilibrium; OGS evaluates the *non-equilibrium initial residual* once,

$$\mathbf{r}^{\text{neq}} = \int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma}_0 \, d\Omega - \mathbf{F}^{\text{ext}}(0). \quad (2.8)$$

Integration by parts shows that for a uniform (divergence-free) initial stress, the free-degree entries of $\mathbf{r}^{\text{neq}}$ live on the excavation boundary Γ_{exc} ; the consistent equivalent nodal force at wall node i is

$$\mathbf{F}_{0,i} = \int_{\Gamma_{\text{exc}}} N_i^T \boldsymbol{\sigma}_0 \mathbf{n} \, d\Gamma. \quad (2.9)$$

For the uniaxial initial stress this support traction has normal component $t_n = -P \sin^2 \theta$ and tangential component $t_\theta = P \sin \theta \cos \theta$; it is a spatially varying traction field, *not* a uniform pressure (that interpretation is valid only for an isotropic initial stress $\sigma_0 = -p_0 \mathbf{I}$).

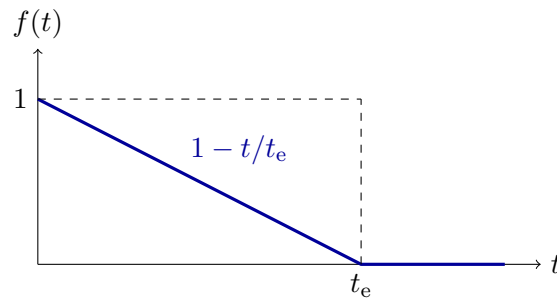
OGS subtracts the stored initial residual in every nonlinear iteration (*residual compensation*), so that $\mathbf{u} = \mathbf{0}$ satisfies equilibrium at $t = 0$; the compensation represents the support of the not-yet-excavated core. The boundary condition then withdraws this support with a user-defined decay function $f(t)$, $f(0) = 1$, yielding the effective wall residual

$$\tilde{\mathbf{r}}_i = \mathbf{F}_i^{\text{int}} - \mathbf{F}_i^{\text{ext}} - f(t) \mathbf{F}_{0,i} = \mathbf{0}. \quad (2.10)$$

The notebook uses the linear decay

$$f(t) = \begin{cases} 1 - t/t_e, & 0 \leq t \leq t_e, \\ 0, & t > t_e, \end{cases} \quad (2.11)$$

so the wall is in-situ supported at $t = 0$, partially unloaded for $0 < t < t_e$, and traction-free for $t \geq t_e$. Because the material is linear elastic, the excavation-induced response scales linearly with the released fraction $1 - f(t)$.



2.5 OGS realization

The process variable requests the residual compensation and the wall boundary condition:

```
<process_variable>
  <name>displacement</name>
  <components>2</components>
  <order>2</order>
  <initial_condition>initial_displacement</initial_condition>
  <compensate_non_equilibrium_initial_residuum>true
</compensate_non_equilibrium_initial_residuum>
  <boundary_conditions>
    <boundary_condition>
      <mesh>arc</mesh>
      <type>ReleaseNodalForce</type>
      <time_decay_parameter>decay_function</time_decay_parameter>
    </boundary_condition>
  </boundary_conditions>
</process_variable>
```

The mechanics process references the initial-stress parameter `sigma0` through its `initial_stress` tag, with the plane-strain components of Eq. (2.2), and the decay function of Eq. (2.11) enters as

```
<parameter>
  <name>decay_function</name>
  <type>Function</type>
  <expression>if (t>172800) 0.0; else 1-t/172800</expression>
</parameter>
```

Two analyses are compared: the *release* formulation (initial stress, `ReleaseNodalForce` wall, compensation enabled) and a *standard* formulation (zero initial stress, natural traction-free wall, load applied in one step from the unstressed configuration). The final total stresses must agree; the displacement fields need not, because they refer to different reference configurations.

2.6 Verification results

The executed notebook (OGS 6.5.9) runs five simulations; all return code zero and all final stress checks are asserted programmatically.

Measure	Analytical	Quadratic	Linear
$\sigma_{rr}(a, 0^\circ)$ / MPa	0	−0.133	−2.620
$\sigma_{\theta\theta}(a, 0^\circ)$ / MPa	−60	−61.581	−62.476
Release–standard maximum stress difference / MPa	0	9.9×10^{-11}	4.5×10^{-12}
Initial release-model displacement / m	0	0	0

The release and standard stress fields agree to numerical precision. The nonzero extrapolated σ_{rr} at the wall is not a failure of the traction boundary condition: OGS computes stress at integration points and extrapolates to nodes for VTK output, and this postprocessing step, together with mesh resolution and element order, creates the wall-node discrepancy. Quadratic elements improve the wall value substantially on the present mesh.

An element-deactivation variant fills the opening with a quarter-disc of inactive elements (material ID 1). OGS assigns zero integration-point stress to inactive cells, so nodal extrapolation at the shared wall nodes mixes physical rock stresses with zeros; the notebook therefore excludes the shared wall node from the checks. Away from that interface, the deactivation and hole-only in-plane stresses agree within 10^{-8} MPa. For wall verification one should use integration-point or cell-based stresses, or a sample point demonstrably inside the active rock.

The notebook additionally demonstrates polar postprocessing (`ot.variables.stress.to_polar`) and maps the mean stress $\sigma_m = \frac{1}{3} \text{tr } \sigma$ and octahedral shear stress $\tau_{\text{oct}} = \sqrt{2J_2/3}$ — the invariant pair relevant for pressure-dependent failure criteria: excavation simultaneously concentrates shear and removes confinement at the wall.

3 Problem 2: Fluid injection into fractured rock (LIE vs. EFPM)

3.1 Purpose and background

The second exercise addresses hydro-mechanically coupled processes in fault-controlled reservoirs, motivated by deep geothermal systems in fractured carbonates. A pre-existing fracture in a nearly impermeable rock matrix is pressurised by fluid injection and subsequently depressurised by production. Two modelling strategies are benchmarked against each other and against analytical references:

- **LIE** — the fracture is a lower-dimensional interface element with an explicit displacement-jump unknown (process `HYDRO_MECHANICS_WITH_LIE`); fracture flow follows the cubic law with pressure-dependent aperture.
- **EFPM** — the fracture is a thin band of standard continuum elements (process `HYDRO_MECHANICS`) whose permeability and stiffness are *equivalent* upscaled properties: the medium property `OrthotropicEmbeddedFracturePermeability` and a Gerrard-type equivalent modulus [6].

Learning objectives: discrete versus equivalent-continuum fracture representations, the cubic law, hydro-mechanical aperture coupling, and a verification methodology combining closed-form and numerical references.

3.2 Model definition

3.2.1 Geometry and mesh

The domain is a plane-strain rock-matrix block of length 12.5 m ($x \in [0.1, 12.6]$ m) and height 1 m. A single horizontal fracture runs along $y = 0.5$ m from the left to the right edge. In the LIE model it is a line of 1D interface elements (34 elements embedded in 1292 quadratic quads); in the EFPM model it is a band of thickness $s = 1$ mm, one element thick (39 band elements within 2301 quads). The double-fracture variant places fractures at $y = 0.3$ m and $y = 0.7$ m. All meshes are structured, transfinite, graded towards the fracture, and converted to quadratic elements; displacement is quadratic, pressure linear (Taylor–Hood).

3.2.2 Fracture flow and the two representations

Fracture flow obeys the *cubic law* [5]: a fracture of hydraulic aperture b has permeability and transmissivity

$$k_F = \frac{b^2}{12}, \quad T_{\text{LIE}} = k_F b = \frac{b^3}{12}. \quad (3.1)$$

In the LIE model the aperture evolves with the normal displacement jump; in linearized form $b = b_0 + \Delta p/K_n$, and the fracture storage combines fluid storage and fracture compliance, $\beta = b S_F + 1/K_n$.

In the EFPM the band of width s carries the upscaled permeability

$$\mathbf{k} = k_M \mathbf{I} + \frac{b}{a} \frac{b^2}{12} (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}), \quad b = b_0 + a \langle \varepsilon_n - \varepsilon_0 \rangle, \quad (3.2)$$

where a is the mean fracture spacing, $\mathbf{n}$ the fracture normal, and ε_n the strain normal to the fracture. Choosing $s = a$ makes the band transmissivity $T_{\text{EFPM}} = k s$ equal to Eq. (3.1). With $b_0 = 0$, the initial aperture $b(t=0) = 10 \mu\text{m}$ is encoded by the threshold strain $\varepsilon_0 = -b(t=0)/a = -10^{-2}$. The band's normal compliance mimics the fracture stiffness through the Gerrard equivalent modulus [6],

$$\frac{1}{E_{\text{RM}}} = \frac{1}{E_M} + \frac{1}{K_n a} \Rightarrow E_{\text{RM}} = 9.98 \times 10^7 \text{ Pa}. \quad (3.3)$$

3.2.3 Parameters

Property	Symbol	Value
<i>Rock matrix</i>		
Young's modulus	E	60 GPa
Poisson's ratio	ν	0
Solid density	ϱ_s	2716 kg/m ³
Porosity	ϕ	0.001
Permeability	k_M	10^{-21} m^2
Biot coefficient	α_B	1
Storage	S	10^{-10} 1/Pa
<i>Fluid</i>		
Density	ϱ_{LR}	1000 kg/m ³
Viscosity	μ	10^{-3} Pa s
Compressibility	κ_f	10^{-7} 1/Pa
<i>Fracture (LIE)</i>		
Normal and shear stiffness	$K_n = K_s$	10^{11} Pa/m
Initial aperture	b_0	10^{-5} m
Fracture storage	S_F	10^{-10} 1/Pa
Permeability model		CubicLawPermeability
<i>Fracture band (EFPM)</i>		
Band thickness, fracture spacing	$s = a$	10^{-3} m
Threshold strain	ε_0	-10^{-2}
Band Young's modulus	E_{RM}	$9.98 \times 10^7 \text{ Pa}$
Fracture normal	$\mathbf{n}$	(0, 1, 0)

3.2.4 Initial conditions, boundary conditions, and injection schedule

The LIE model works with absolute quantities: initial and far-field pore pressure $p_0 = 11 \text{ MPa}$ (also a Dirichlet condition at the outlet point on the right edge), vertical total load $\sigma_{yy} = -50 \text{ MPa}$ on the top boundary, and an initial effective stress of $\sigma'_{yy} = -39 \text{ MPa}$ in the matrix and on the fracture. The EFPM model is formulated in *relative* quantities (zero initial pressure, load, and stress); its results are compared after adding p_0 . Displacement conditions are $u_y = 0$ on the bottom and $u_x = 0$ on the left and right boundaries.

Fluid is injected at the left fracture end at the mass rate $\dot{m} = 2 \times 10^{-5} \text{ kg}/(\text{s m})$, i.e. a volumetric rate $Q = 2 \times 10^{-8} \text{ m}^3/(\text{s m})$: in the LIE model as a point source at the fracture inlet node, in the EFPM model as the equivalent flux $Q/s = 2 \times 10^{-2} \text{ m/s}$ on the 1 mm inlet edge of the band. The schedule ramps the rate up over 0–100 s, holds it constant until 1000 s, reverses it linearly over 1000–1100 s (switch from injection to production), and holds the reversed rate until the end time $t_{\text{end}} = 1500 \text{ s}$.

3.2.5 Numerical settings

Backward Euler with iteration-number-based time stepping (initial $\Delta t = 10 \text{ s}$, $\Delta t \in [0.1, 20] \text{ s}$); damped Newton (damping factor 0.8, maximum 50 iterations) with per-component Δx criteria; Eigen SparseLU with scaling as the linear solver. All three course cases run with the stock OGS 6.5.9 wheel: single-fracture LIE in 83 s (76 steps), single-fracture EFPM in 357 s (95 steps), double-fracture EFPM in about 3 minutes (76 steps), with no rejected steps. The systems are ill-conditioned (the $K_n = 10^{11} \text{ Pa/m}$ penalty-like fracture stiffness), which is why a direct solver is used; on builds with Intel MKL, PardisoLU solves the same systems roughly $2.8\times$ faster.

3.3 Verification and results

3.3.1 Analytical and 1D references

Two independent references are built in the notebook for the single-fracture case:

1. a *linearized closed-form solution* of the fracture diffusion problem of Carslaw–Jaeger type [7] (constant-flux half-line, `ierfc` kernel), with the 100 s injection ramp incorporated exactly via Duhamel's principle; and
2. a *nonlinear 1D finite-difference solution* of

$$\beta(b) \frac{\partial p}{\partial t} = \frac{\partial}{\partial x} \left(\frac{b^3}{12\mu} \frac{\partial p}{\partial x} \right), \quad b = b_0 + \frac{p - p_0}{K_n}, \quad \beta = b S_F + \frac{1}{K_n}, \quad (3.4)$$

which retains the cubic-law nonlinearity and the ramped source $Q(t)$.

Including the injection ramp is essential: without it both references lead the simulations systematically. The initial transmissivity, storativity, and hydraulic diffusivity are $T_0 = 8.33 \times 10^{-14} \text{ m}^3/(\text{Pa s})$, $\beta_0 = 1.00 \times 10^{-11} \text{ m/Pa}$, and $D_0 = 8.33 \times 10^{-3} \text{ m}^2/\text{s}$.

3.3.2 Key results

Quantity (single fracture)	LIE	EFPM	1D reference
Peak inlet pressure / MPa	11.489	11.480	—
Inlet overpressure at $t = 1000 \text{ s}$ / kPa	489.0	—	505.1
Inlet pressure at $t = 1500 \text{ s}$ / MPa	10.808	10.795	—
Maximum $ u_y $ at $t \approx 1000 \text{ s}$ / μm	4.79	4.69	—

- **LIE vs. EFPM.** Inlet and outlet pressure histories and pressure profiles along the fracture agree within about 0.1–0.2 % throughout injection and production; displacement fields agree within a few percent. The equivalent-continuum representation therefore reproduces the discrete-fracture response for this configuration.

- **LIE vs. analytical references.** The simulated inlet overpressure at the end of injection deviates by 3.3 % from the nonlinear 1D reference; the residual difference is the 2D elastic compliance of the matrix, which the 1D references do not contain.
- **Aperture coupling.** During injection the inlet aperture opens from 10 to about $14.7 \mu\text{m}$ (fracture permeability from 1.0×10^{-11} to $1.8 \times 10^{-11} \text{ m}^2$, consistent with $b^2/12$ — i.e. the transmissivity roughly triples), while the outlet aperture stays at $10 \mu\text{m}$; during production the inlet aperture closes below its initial value, to about $8.8 \mu\text{m}$ at $t = 1500 \text{ s}$. The pressure pulse is still propagating outward along the fracture while the inlet is already depressurising.
- **Double fracture (EFPM).** Both fractures respond nearly identically to each other and to the single-fracture case (peak inlet pressures within 0.01 MPa); the maximum vertical displacement roughly doubles to about $9.4 \mu\text{m}$ because both fractures dilate.

As noted in Section 1.2, released OGS versions support only one displacement-jump variable, so the double-fracture configuration is run with the EFPM approach in the course; a double-fracture LIE project file is included for participants with access to the development branch.

4 Problem 3: Heated tunnel in Opalinus Clay (TRM)

4.1 Purpose and background

The third exercise models the near field of a heat-emitting tunnel — a plane-strain abstraction of the FE heater experiment at the Mont Terri rock laboratory — with the OGS `THERMO_RICHARDS_MECHANICS` (TRM) process. It follows the systematic benchmark of Pitz et al. [2] (DECOVALEX-2023 Task C); the reference paper is included in the exercise folder. Learning objectives:

- set up a coupled THM model under the Richards assumption;
- understand thermal pressurisation of pore water in a low-permeability host rock and its anisotropic drainage;
- model unsaturated behaviour: van Genuchten retention and relative permeability, Bishop effective stress, vapour diffusion, latent heat, saturation-dependent thermal conductivity;
- use orthotropic materials with local coordinate systems; and
- work with mixed-order (Taylor–Hood) elements and interpolated output fields.

4.2 Model definition

4.2.1 Geometry and mesh

A quarter of a $50\text{ m} \times 50\text{ m}$ domain is discretised (quarter dimensions $25\text{ m} \times 25\text{ m}$). A granular bentonite buffer ring (material ID 0) extends from the heater surface at $r_i = 0.525\text{ m}$ to the tunnel wall at $R = 1.24\text{ m}$; the surrounding Opalinus Clay is material ID 1. The heater itself is not meshed; it acts through a heat-flux boundary condition on the inner arc.

The mesh is a structured, transfinite quadrilateral Gmsh mesh (15 nodes per block edge, 8 radial nodes across the buffer), converted to 8-node quadratic quadrilaterals: 1176 cells (196 buffer, 980 clay) and 3683 nodes. The displacement is interpolated with quadratic shape functions and pressure and temperature with linear ones (Taylor–Hood). Boundary meshes `left/right/top/bottom/arc_inner/arc_outer` are extracted automatically.

4.2.2 Process and governing equations

The TRM process solves, in the domain Ω and for the unknowns temperature T , liquid pressure p_{LR} , and displacement $\mathbf{u}$:

- the **energy balance** of the fluid–solid mixture with effective volumetric heat capacity $(\rho c_p)_{\text{eff}}$, heat conduction with the effective conductivity λ_{eff} , and advection by the Darcy flux (plus vapour-related latent-heat transport in the unsaturated configuration);

- the **liquid mass balance** under the Richards assumption (gas at constant atmospheric pressure), with storage from fluid and pore compressibility, saturation changes $\phi \partial S_L / \partial t$, thermal expansion of the pore water, Darcy flow $\mathbf{q} = -\frac{k_{\text{rel}}(S_L) \mathbf{k}}{\mu(T)} \nabla p_{\text{LR}}$, and, in the unsaturated configuration, diffusive water-vapour transport; and
- the **momentum balance** $\nabla \cdot (\boldsymbol{\sigma}' - \alpha_B \chi(S_L) p_{\text{LR}} \mathbf{I}) = \mathbf{0}$ with the Bishop factor χ and thermo-elastic effective stress $\boldsymbol{\sigma}' = \mathbb{C} : (\boldsymbol{\varepsilon} - \alpha_T \Delta T \mathbf{I})$.

Gravity is disabled. With $S_L \equiv 1$, $k_{\text{rel}} \equiv 1$, and $\chi = S_L$ the formulation reduces to single-phase saturated THM — this is the *saturated* configuration (TRM-S in [2]). The *partially saturated* configuration activates van Genuchten saturation and relative permeability, a Bishop saturation-cutoff model, water-vapour density and FEBEX-type vapour diffusion with linear latent heat, mass lumping, and a temperature-dependent IAPWS viscosity curve.

4.2.3 Material parameters

Property	Symbol	Bentonite buffer	Opalinus Clay
Solid density / kg m^{-3}	ρ_s	2227.2	2689.66
Solid specific heat / $\text{J kg}^{-1} \text{K}^{-1}$	$c_{p,s}$	800	995
Solid thermal conductivity / $\text{W m}^{-1} \text{K}^{-1}$	λ_s	1.2	$\lambda_{\perp} = 1.3$, $\lambda_{\parallel} = 2.4$
Solid thermal expansivity / K^{-1}	α_T	3.0×10^{-6}	1.5×10^{-5}
Intrinsic permeability / m^2	k	3.5×10^{-20}	$k_{\perp} = 10^{-20}$, $k_{\parallel} = 5 \times 10^{-20}$
Porosity	ϕ	0.331	0.13
Elasticity		isotropic, $E = 18 \text{ MPa}$, $\nu = 0.35$	orthotropic, see text
Biot coefficient	α_B	1.0	1.0
van Genuchten p_b, m		28.6 MPa, 0.5	20 MPa, 0.6

The clay is modelled as a linearly elastic orthotropic material with $E_{\perp} = 4 \text{ GPa}$ and $E_{\parallel} = 8 \text{ GPa}$, Poisson's ratios (0.25, 0.35, 0.35), and shear moduli $G = 3.5 \text{ GPa}$ and $8/(2 \cdot 1.35) \approx 2.96 \text{ GPa}$; its anisotropy axes are set through an OGS local coordinate system (`<local_coordinate_system>`), with bedding-parallel and bedding-perpendicular directions as listed above. Two saturated bedding orientations (0° and 90°) are generated from the same parameter set by rotating the local basis and the permeability tensor.

The pore water uses $c_p = 4184 \text{ J/(kg K)}$, $\lambda = 0.63 \text{ W/(m K)}$, viscosity $\mu = 10^{-3} \text{ Pa s}$, and a linear equation of state $\rho_{\text{LR}}(p, T)$ with reference density 1000 kg/m^3 , thermal expansivity $4 \times 10^{-4} \text{ 1/K}$, and compressibility $5 \times 10^{-10} \text{ 1/Pa}$. The partially saturated configuration replaces the constant viscosity by a 37-point IAPWS $\mu(T)$ curve (273.15–453.15 K) and uses a buffer thermal conductivity that increases linearly with saturation from 0.35 (dry) to 1.2 W/(m K) (saturated).

4.2.4 Initial and boundary conditions

Field	Saturated configuration	Partially saturated configuration
T_0	288.15 K (15 °C)	288.15 K
p_0	0 Pa (results shifted by +2 MPa in the plots)	buffer -123 MPa ($S_L \approx 0.23$, dry emplacement); clay +2 MPa
u_0, σ_0	zero	zero
Displacement	rollers on all four outer boundaries ($u_x = 0$ left/right, $u_y = 0$ top/bottom); heater surface (<code>arc_inner</code>) fixed	
Temperature	Neumann heat flux $q = 89 \text{ W/m}^2$ on <code>arc_inner</code> , constant over the full five years	
Hydraulic	all boundaries natural (no flow)	

Because the saturated response is linear in the pressure level, running it with $p_0 = 0$ and shifting the plotted pressures by +2 MPa reproduces the paper's absolute values exactly.

4.2.5 Numerical settings

Backward Euler in time with iteration-number-based adaptive time stepping (initial $\Delta t = 0.1 \text{ s}$, maximum $\Delta t = 1.576 \times 10^6 \text{ s}$) up to $t_{\text{end}} = 1.5768 \times 10^8 \text{ s}$ (5 years). Newton's method (maximum 20 iterations) with a per-component relative Δx criterion of 10^{-6} for T, p, u_x, u_y . The linear solver is Eigen's `SparseLU` direct solver with equilibration (`<scaling>true</scaling>`); iterative presets (BiCGSTAB/GMRES with ILUT) are available in the notebook as named alternatives. Because T and p are first-order fields on a second-order mesh, the project outputs `temperature_interpolated` and `pressure_interpolated`, which carry values on *all* nodes of the quadratic mesh; all probes and contour plots use these fields.

4.3 Results

All three runs (saturated 0°, saturated 90°, partially saturated) complete with OGS 6.5.9 without any rejected time steps (139/139/142 adaptive steps; about 80–150 s wall time each). Representative values at the end of the five-year heating phase, probed on the horizontal symmetry axis (saturated pressures include the +2 MPa shift):

	Heater surface $r = 0.525 \text{ m}$	Tunnel wall $r = 1.24 \text{ m}$	Clay $r = 3 \text{ m}$
<i>Saturated (90° bedding case)</i>			
Temperature	123.9 °C	84.4 °C	63.4 °C
Liquid pressure	18.5 MPa	18.4 MPa	17.4 MPa
<i>Partially saturated</i>			
Temperature	161.7 °C	79.8 °C	54.4 °C
Liquid pressure	-478 MPa	-32.4 MPa	+2.5 MPa
Saturation	0.06	0.52	1.00

Saturated configuration. Heating pressurises the pore water of the low-permeability clay to a peak of about 16.5 MPa above the initial state (thermal pressurisation), which then drains anisotropically along the bedding. The two bedding orientations 0° and 90° produce stress, pressure, and displacement fields that are exact 90° rotations of each other (maximum displacement 6.08 mm in both runs), a built-in consistency check of the local coordinate system and tensor rotation handling.

Partially saturated configuration. The dry-emplaced buffer first desiccates further near the heater (saturation drops from 0.23 to about 0.06, suction grows to several hundred MPa) because the strong thermal gradient drives vapour outward, while the buffer simultaneously resaturates from the host rock at the tunnel wall. The clay pressurises to a peak of about 11 MPa. The heater temperature is markedly higher than in the saturated case ($\approx 162^\circ\text{C}$ vs. $\approx 124^\circ\text{C}$) because the dry buffer conducts heat poorly; as the buffer resaturates its conductivity rises and the heater temperature declines late in the simulation. This is the characteristic behaviour reported in [2]; digitized reference curves from the paper are archived in `TRM/paper_data/` for further comparison exercises.

Mesh sensitivity. A fine-mesh reference solution (5700 cells, archived in the folder `TRM/_out_fine_reference/`) confirms the saturated fields on the course mesh to within a fraction of a degree and 0.1 MPa. The steep dry-out front of the partially saturated case is more mesh-sensitive: the fine mesh gives 151°C and -380 MPa at the heater against 162°C and -478 MPa on the coarse course mesh. Participants should treat near-heater suction values as mesh-dependent and refine locally when they matter.

5 Concluding remarks

The three exercises cover the span from single-physics verification to multi-field, multi-material applications:

- Problem 1 establishes the methodology: a formal derivation of the excavation boundary condition, exact agreement between two formulations, and a quantitative comparison with a classical analytical solution — including the discipline of separating model error, discretization error, and postprocessing artifacts.
- Problem 2 contrasts two fracture representations inside one verification framework and shows that an equivalent-continuum model, when its transmissivity and compliance are upscaled consistently, matches a discrete interface-element model to within a fraction of a percent.
- Problem 3 scales this methodology to a coupled THM problem with two materials, anisotropy, and unsaturated constitutive behaviour, reproducing the published heater benchmark behaviour and including a mesh-sensitivity assessment.

All folders are self-contained and re-runnable from scratch with the single Python environment of Section 1.2; the executed course versions were verified with OpenGeoSys 6.5.9.

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